WEAPONS ROBOTICS AND CONTROL ENGINEERING

HOPPER INFORMATION SERIES

OBJECT DETECTION

Object detection and COCO

How Hopper Studio turns camera pixels into labeled boxes with a host-side neural network.

COCO-SSD

MOBILENETV2

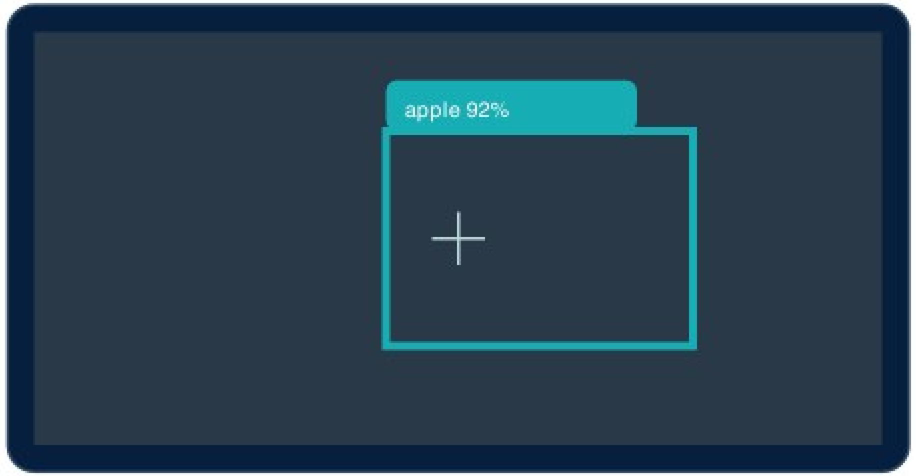
BOUNDING BOXES

WHAT QUESTION IS ANSWERED?

Detection is not the same as classification

OBJECT DETECTOR

Find multiple known objects and report where each one is.



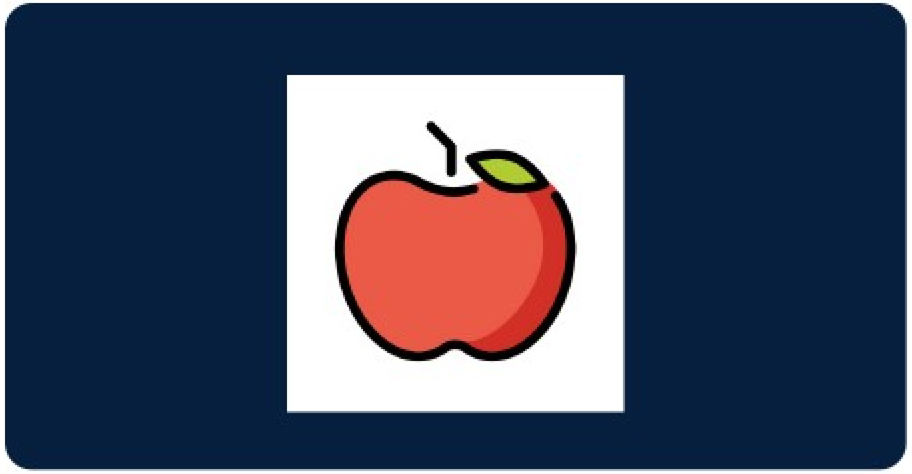
OUTPUT

`class + confidence + box + x/y`

Built-in COCO-SSD

IMAGE CLASSIFIER

Choose the best whole-frame class. It does not draw a box.



`apple landing pad  91%`

Custom Teachable Machine

COCO: common objects pictured in real context

330K
IMAGES

>200K
LABELED IMAGES

1.5M
OBJECT INSTANCES

80
OBJECT CATEGORIES

Why “in context” matters

• Scenes contain multiple objects, clutter, overlap, and non-iconic viewpoints.
• Annotations include bounding boxes and segmentation masks—not only image labels.
• The 80 detector classes cover everyday categories such as person, car, truck, stop sign, chair, laptop, bottle, and sports ball.

What COCO does not guarantee

• A class outside the 80-label vocabulary cannot be named.
• A familiar class may still be missed when it is tiny, blurred, occluded, or viewed unlike the training data.
• A confidence score is a model score—not certainty.

Dataset scale creates breadth; classroom testing reveals the limits.

DEFAULT NEURAL NETWORK

Inside the shipped COCO-SSD detector

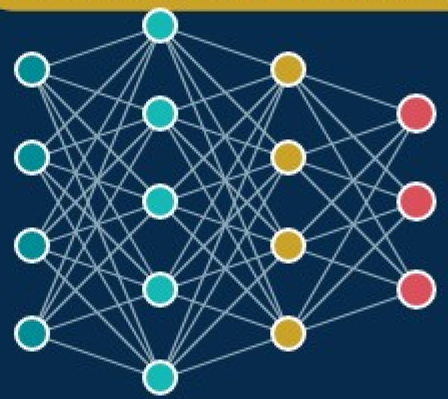
HOST-SIDE · NOT ONBOARD THE QUAD

CAMERA PIXELS

Canvas capture
≤ 420 px wide
[1, H, W, 3] RGB

SSDLITE MOBILENETV2 300

INTERNAL TENSOR · 300×300×3



Depthwise + inverted-residual
convolutional feature extractor

SIX FEATURE MAPS

19×19×576	3 anchors/cell
10×10×1280	6 anchors/cell
5×5×512	6 anchors/cell
3×3×256	6 anchors/cell
2×2×256	6 anchors/cell
1×1×128	6 anchors/cell

BOX + CLASS HEADS

1,917

CANDIDATE ANCHORS

Each anchor predicts

4 box offsets + 91 logits

JavaScript NMS → ≤10 boxes
80 named COCO classes

300×300×3
INTERNAL INPUT TENSOR

4,500,927
COEFFICIENTS / 18.03 MB

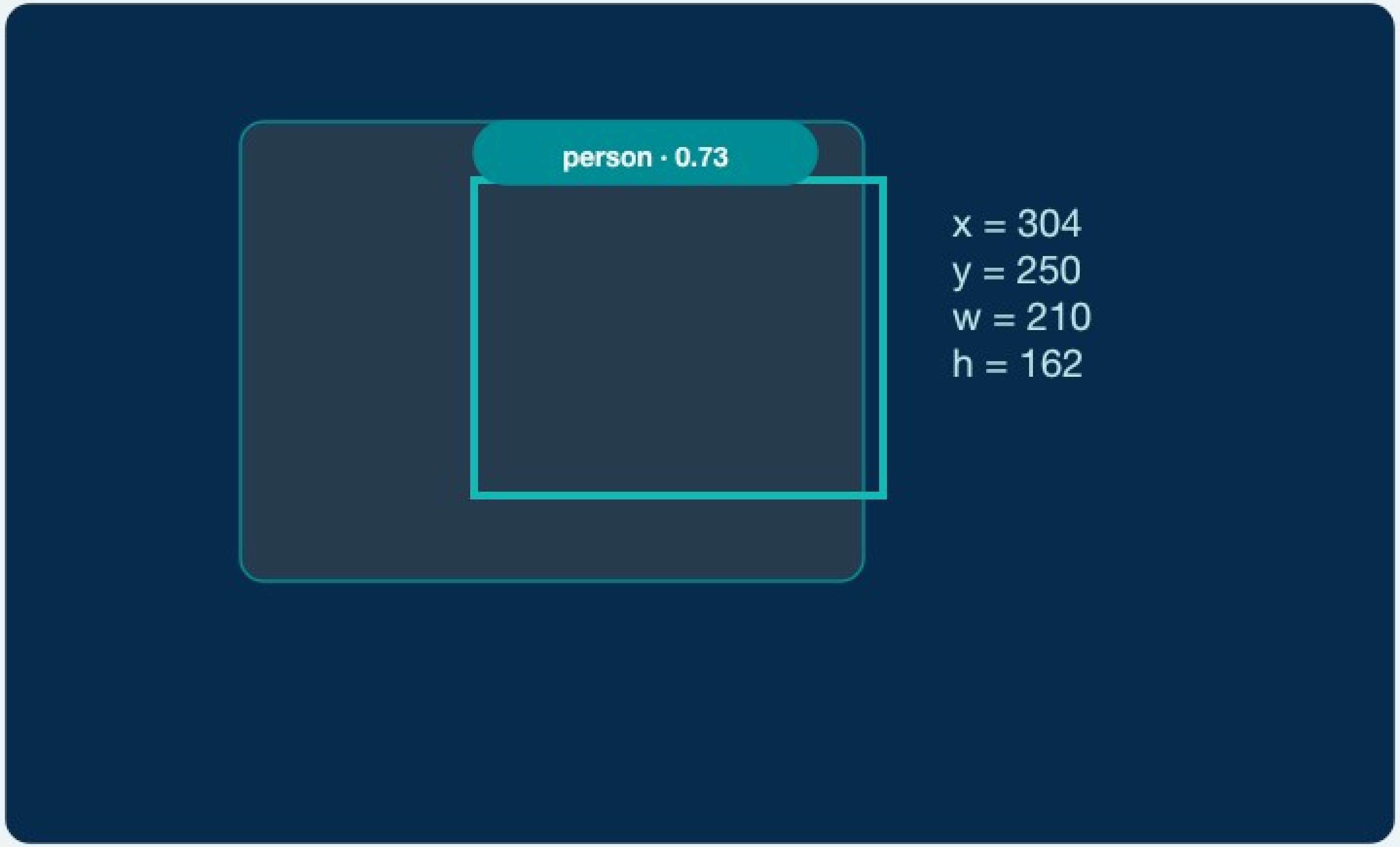
1,917
ANCHORS ACROSS 6 MAPS

0.55
DEFAULT SCORE THRESHOLD

NO MODEL
SIMULATOR: SYNTHETIC BOXES

DECISION THRESHOLD

Confidence controls which boxes become actionable



If the threshold is…

- 0.50 → this box is returned.
- 0.75 → this box is filtered out.
- Lower thresholds show more candidates and more false alarms.
- Higher thresholds show fewer, stronger candidates and more misses.

Coordinate rule

- Bounding boxes use [x, y, width, height].
- Hopper Studio stores the box center as a signed percentage from frame center.

Tune confidence on the exact room, target size, camera angle, and lighting.

MISSION CODE

Use fresh detections and document every argument

FRESH DETECTION DECISION · PROGRAM.JS

```
await drone.takeOff();
try {
  const detections = await vision.detectObjects(0.65);
  const person = detections.find(
    (item) => item.class === "person",
  );

  if (person && person.centerX > 10) {
    await drone.fly("right", 0.4, 12);
  }
} finally {
  await drone.land();
}
```

OBJECT DETECTOR

```
await vision.detectObjects(minimumConfidence, announceScan)
```

minimumConfidence — 0…1 score; default 0.55
announceScan — boolean; default true
returns — Promise<VisionDetection[]>; at most 10 boxes
fields — class, score, bbox, frame size, centerX/centerY
throws — if local model or camera pixels are unavailable
note — simulator returns synthetic detections

FLIGHT COMMAND USED ABOVE

```
await drone.fly(direction, seconds, power)
```

direction — "right" here; six direction strings are accepted
seconds — number ≥ 0; example 0.4 s
power — −100…100%; example 12
returns — Promise<void>; await before landing

MODEL CHOICE

Built-in detector or custom classifier?

BUILT-IN COCO-SSD DETECTOR

• Finds multiple objects and draws a box around each.
• Returns one of 80 fixed COCO labels.
• No training required; useful for people, vehicles, stop signs, laptops, and more.
• Choose this when location or multiple objects matter.

CUSTOM TEACHABLE MACHINE CLASSIFIER

• You choose mission-specific labels: red flag, blue flag, safe pad, obstacle.
• Fast transfer learning with your own classroom examples.
• Returns probabilities for the whole frame—not bounding boxes.
• Choose this when the frame's dominant class answers the mission question.

Need custom labels AND boxes?

That is a custom object detector: collect box-annotated images and train a detection architecture. Hopper Studio's Teachable Machine path is classification.

EXPERIMENT DESIGN

Treat every detection as a measurement with limits

MAKE COCO EASIER

- Use a real label from the 80-class list.
- Choose large, well-separated objects.
- Test distance, rotation, clutter, and background.
- Keep confidence aligned with false-positive and miss costs.

KNOWN FAILURE MODES

- Small or partly hidden objects.
- Motion blur and low light.
- Unusual viewpoints and look-alike classes.
- A target that simply is not in COCO's vocabulary.

Recommended sources

COCO dataset · TensorFlow.js COCO-SSD · SSD paper · MobileNetV2 paper · Hopper Studio source

SCAN → FILTER → LOCATE → DECIDE